Project Title: Malaria Detection Using Deep Learning (CNN) **Project Design Phase-I - Solution Fit Template** **Team ID: 699ee179f292113d819d8e0e**

Define CS, fit into CC

1. CUSTOMER SEGMENT(S) CS

Who is your customer?
i.e. working parents of 0-5 y.o. kids

Primary users of the malaria detection system include:

- Doctors and healthcare professionals working in hospitals
- Laboratory technicians analyzing blood smear images
- Diagnostic centers performing malaria testing
- Healthcare researchers studying AI-based disease detection
- Rural healthcare workers in remote areas with limited diagnostic facilities

6. CUSTOMER CONSTRAINTS CC

What constraints prevent your customers from taking action or limit their choices of solutions? i.e. spending power, budget, no cash, network connection, available devices.

Challenges faced by users include:

- Limited availability of trained pathologists
- Lack of advanced diagnostic equipment in rural areas
- Limited computational resources in small clinics
- Internet connectivity issues in remote healthcare centers

The system should be simple and easy to deploy.

5. AVAILABLE SOLUTIONS AS

Which solutions are available to the customers when they face the problem
or need to get the job done? What have they tried in the past? What pros &

Existing malaria detection methods include:

- Manual microscopic examination of blood smear slides
- Rapid diagnostic test (RDT) kits
- Laboratory-based diagnostic systems

However, these solutions may be slow, expensive, or require expert knowledge.

Explore AS, differentiate

Focus on J&P, tap into BE, understand RC

2. JOBS-TO-BE-DONE / PROBLEMS J&P

The main problem addressed by this project is the slow and manual process of malaria diagnosis.

Key problems include:

- Manual microscopic examination of blood smear images is time consuming
- Shortage of trained pathologists in rural areas
- Human error during manual diagnosis
- Large number of samples to analyze in hospitals

The system aims to automate this process using artificial intelligence.

9. PROBLEM ROOT CAUSE RC

The root cause of the problem includes:

- Lack of automated diagnostic tools
- Dependence on manual microscopic analysis
- Shortage of trained medical professionals
- High workload in hospitals and laboratories

Artificial intelligence can help reduce these challenges.

7. BEHAVIOUR BE

Challenges faced by users include:

- Limited availability of trained pathologists
- Lack of advanced diagnostic equipment in rural areas
- Limited computational resources in small clinics
- Internet connectivity issues in remote healthcare centers

The system should be simple and easy to deploy.

Focus on J&P, tap into BE, understand RC

Identify strong TR & EM

3. TRIGGERS TR

Triggers that motivate users to adopt the system include:

- Increasing number of malaria cases in developing countries
- Need for faster diagnostic methods
- Availability of AI-based medical technologies
- Hospitals adopting automated healthcare solutions
- Government and research initiatives promoting digital health systems

10. YOUR SOLUTION SL

The proposed solution is an **AI-powered malaria detection system** using deep learning.

Key features:

- Convolutional Neural Network (CNN) model for image classification
- Automatic detection of parasitized and uninfected blood cells
- Web application interface built using Flask
- Upload microscopic images and get instant predictions

8.CHANNELS of BEHAVIOUR CH

ONLINE

- Hospital information systems
- AI diagnostic platforms
- Medical research websites

	After using the system: • Doctors feel confident and supported by AI tools • Faster diagnosis improves decision making • Patients receive quick and reliable results	• Improves diagnostic speed and accuracy This system provides an efficient and scalable solution for automated malaria diagnosis.	OFFLINE • Hospitals and clinics • Diagnostic laboratories • Healthcare camps in rural areas	Identify strong TR & EM